An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka

R26-DS-004



Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in
Data Science

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Sri Lanka Institute of Information Technology
Sri Lanka

March 2026

Personalized Recommendation & Predictive Impact Modeling Engine.

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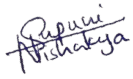
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Annual income taxation requires salaried individuals to file returns each year; however, it is hard to execute good tax savings plans because of different levels of income, financial behaviour, risk tolerance and liquidity constraints. Tax advisers typically provide rule based (i.e., fair and equitable taxation) calculations for preparing and filing income taxes or using static optimisation methodologies in computing optimum taxation strategies. However, while these methodologies identify tax strategies to be technically optimum (meaning that they provide the lowest amount of tax liability), they do not evaluate if taxpayers would actually implement these strategies or how the implementation of these strategies would impact their future financial situation. Therefore, many of the tax strategies that are recommended by tax advisers do not exist in the real world. This results in taxpayers failing to use these strategies and missing out on the long-term benefits of the tax strategies.

To address this limitation, this research proposes a **Personalized Recommendation and Predictive Impact Modeling Engine** that generates behavior-aware tax recommendations and estimates their potential long-term financial impact. The system evaluates multiple tax strategies generated based on Sri Lankan tax regulations and ranks them using a multi-objective scoring framework that considers expected tax savings, adoption likelihood, financial feasibility, and associated risk. An adoption probability model predicts whether a user is likely to implement a recommended strategy by analyzing factors such as income level, financial flexibility, behavioral patterns, and risk tolerance.

A learning-to-rank model like LambdaMART will rank tax strategies for the best fit and effectiveness per user. The tax strategies will be based on a salary forecast model and a Monte Carlo simulation to project long-term financial results of adopting different strategies, including future tax obligations and potential savings.

The design of this system will use Python based machine learning frameworks such as Scikit-learn, Pandas, and NumPy to create synthetic financial datasets and user behavior data via simulation due to privacy constraints with implementing the project. The system will provide a decision-support dashboard that ranks recommendations and shows financial projections for recommendations in an easily understood format.

The project plans to produce an intelligent tax decision-support system that will improve tax strategy adoption; increase compliance; and provide transparent insight into the long-term financial effects of tax decisions.

Keywords: Personalized Tax Recommendation, Learning-to-Rank, Adoption Modeling, Predictive Impact Modeling, Monte Carlo Simulation.

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LIST OF ABBREVIATIONS

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **IRD** – Inland Revenue Department
- **APIT** – Advance Personal Income Tax
- **DSR** – Design Science Research
- **SDG** – Sustainable Development Goals
- **USP** – Unique Selling Proposition
- **API** – Application Programming Interface
- **NDCG** – Normalized Discounted Cumulative Gain
- **MAP** – Mean Average Precision
- **GPU** – Graphics Processing Unit
- **OECD** – Organisation for Economic Co-operation and Development

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1. INTRODUCTION

1.1 Background and Context

1.1.1 Real -World Problem

Income taxes are a key component of the government's revenue stream, as well as help to provide for stability within an economy and maintain overall economic growth. Individuals in Sri Lanka are required to complete an annual income tax return, according to the provisions of the Inland Revenue Act No.24 of 2017. The majority of salaried employees have income tax withheld directly from their salaries via the Advance Personal Income Tax (APIT), but taxpayers still must report directly to the Inland Revenue Department (IRD) their total income and tax liabilities each year.

When planning for taxes, many taxpayers struggle to find effective methods of reducing their taxes by finding appropriate tax-saving strategies that fit their own financial situations. Various factors will need to be assessed by a taxpayer when planning for their taxes, including: income level; eligible deductions; available types of investments; and any limitations imposed by laws and/or other regulations. The complexity of tax regulations and changes in tax laws make it challenging for many taxpayers to determine the most appropriate tax-saving strategies.

Taxpayers often rely on either standard tax calculators or generic financial advice that does not take into account their specific financial circumstances. This can lead to taxpayers making poor choices regarding their financial future. The majority of taxpayers who would benefit from effective tax planning are unable to find the necessary information and guidance to do so.

Thus, there may be a mismatch between the ideally efficient tax strategy as contemplated by theory versus the actual tax strategies used in practice by taxpayers. This sometimes creates inefficiencies in the financial planning process and results in taxpayers missing opportunities to save taxes legitimately. To solve this problem, it is necessary to provide intelligent decision-support systems that will supply individuals with accurate/legitimate, practical suggestions regarding their individual tax situations

1.1.2 Challenges Faced by Taxpayers in Selecting Tax Strategies

Many taxpayers have difficulty choosing the best way to minimize their tax bill because there are many different ways to reduce taxes legally. One of these ways is through tax regulations; tax laws (and therefore their tax liability) can be complex and therefore challenging to assess how to take advantage of any tax incentive policies. In Sri Lanka, there are multiple deductions, exemptions, and tax relief/benefits related to investments on the basis of the Inland Revenue Act. For many taxpayers, the difficulty lies in understanding which of the many different rules they can utilise based on their specific financial circumstances and which are applicable to them [1], [2].

One significant impediment to optimizing tax strategy is poor financial knowledge & individualised assistance by taxpayers. Many rely on the same calculators to determine their estimated taxes as those who use the most simplistic advice from general sources. Therefore, many taxpayers find it challenging to evaluate various ways to decrease their taxes; consequently are unable to develop an optimal plan that considers individual circumstances such as level of income and risk tolerance, as well as possible financial solutions. Research indicates that intelligent decision support systems will enhance the quality of tax planning & financial decision making [5].

Similarly, differences in financial behaviours and risk profiles pose additional challenges for tax payers. Some people are more inclined to take risk, make a long-term investment or engage in a structured saving programme than others are. Several traditional tax consulting systems utilize only theoretical tax savings to develop an optimal tax strategy; consequently, these tax saving strategies may or may not be achievable by the taxpayer [6].

One major challenge is the lack of predictive capabilities to evaluate the future financial impact of tax decisions. While many existing tax advisory tools calculate current or immediate tax amounts, they do not offer insight into how different strategies might affect long-term financial results, such as future tax liabilities and savings potential. Outside of the world of taxation, predictive analytics and simulation models such as Monte Carlo simulation have been used successfully to enhance financial forecasting and provide decision support in complicated financial scenarios [13].

In addition, there are currently no existing models designed to rank tax strategies based on the user's preferences, ability to implement and likelihood of ongoing use. Learning-to-rank models have been commonly used in contemporary recommendation systems to order the most appropriate options for users based the success criteria [7],[8].

1.1.3 Statistical Evidence (Global and Research Evidence)

Artificial Intelligence (AI) and other Technology Advancements have Enhanced the Quality of Decision Support Systems and Financial Analysis. There is a greater Increase in the use of machine learning systems within Taxation Systems to analyze financial records, provide supporting documentation for tax compliance and advisory services, and offer solutions. Research has demonstrated that Machine Learning systems improve tax compliance monitoring and facilitate better risk analysis and decision making by utilizing large volumes of data to identify patterns in financial records and provide predictive analysis to both taxpayers and government regulatory authorities [5],[6].

Recommendation systems have been widely adopted across many industries (finance, healthcare, ecommerce...) to give individual users personalized decision support. These systems analyze user behavior and preference histories to recommend the best course of action for each individual. Specifically, learning-to-rank algorithms are commonly utilized as they provide decision-makers with the right alternatives based on multiple decision-making attributes (e.g., relevance, expected benefit, and user preference [7],[8].

Furthermore, the use of predictive modeling methods is widely accepted in managing financial resources for estimating long-term results of financial decision making. Simulation techniques (e.g., Monte Carlo analysis) allow the financial management process to evaluate many potential scenarios and the consequences of implementing a variety of different financial strategies under uncertain/economic conditions.These techniques have been widely applied in financial planning and investment analysis to estimate future financial risks and benefits [13].

1.1.4 Analysis of Existing Solutions and Their Limitations

To help taxpayers calculate their tax obligation, and to file their tax return, a variety of digital tax advisory tools and financial systems have been developed. The Inland Revenue Department in Sri Lanka provides an online platform and official guidelines to assist taxpayers with tax regulation and the filing of annual income tax returns.
While there is greater access and therefore more ease for taxpayers to file their taxes because of the systems developed, these systems mainly focus on compliance and calculation of taxes, not support in choosing an appropriate tax savings strategy.

Most of the existing tax systems are based on rule-based tax calculations, where the computation of taxpayer obligations are based on predetermined pre-defined calculations and deduction rules. This ensures that taxpayers comply with existing regulations; however, the absence of a personalized recommendation based on the individual's financial situation and long-term financial plan limits or eliminates what taxpayers are able to implement.

Advisory tools tend to only focus on maximizing the tax savings that a taxable person can achieve, and do not take into consideration other practical considerations such as financial flexibility, liquidity constraints or risk tolerance. Because of this, some of the recommended tax-savings strategies may not be practical for taxpayers to implement.

When designing tax strategy recommendation systems, user behavior and probability of adopting a given strategy is generally not taken into consideration by most current solutions. Studies have shown that incorporating machine learning (ML) techniques into financial decision support systems can lead to an increase in the effectiveness of recommendations and improved financial decision making [5] [6].

In addition, most tax advisory platforms do not assess how a tax decision will affect that individual's long-term financial situation. Some methods (e.g., financial forecasting and Monte Carlo simulations) exist to provide estimates of a given financial situation at various points in time, but they are infrequently used in tax advisory systems [13].

Many modern recommendation systems use learning to rank methods to create a list of all the possible recommendation options for the user by ranking them according to the probability of selecting that option. However, these methods are seldom used in tax strategy recommendation engines [7] [8].

1.1.5 Proposed Solution and Its Key Components

This research introduces an innovative Personalized Recommendation and Predictive Impact Modelling Engine designed for enhancing tax decisions made by individual taxpayers. The primary purpose of the developed system is to generate personalized recommendations for saving taxes for the user's current financial situation as well as their behavioural preferences, and estimated long-term financial outlook.

The proposed system contains four fundamental components:

1) **User Financial Profile Module** – The User Financial Profile Module gathers and analyses user-specific financial data including the individual's income level; salary history; deductions eligible for tax; investment preferences; and their individual risk tolerance to develop a constructed financial profile that is representative of the user's financial behaviour and decision-making characteristics.

2) Tax Strategy Generation Module – The Tax Strategy Generation Module generates multiple different possible tax-saving strategies (that may be applicable to the user based on their User Financial Profile) based on the various tax rules established by the Sri Lankan Government and their various deductions that are available at that time. The tax-saving strategies that may be produced can include different types of deductions, savings plans to reduce tax owed, and/or forms of tax-eligible financial activity.

3) Recommendation Ranking Engine - This module would assess and rank the tax planning strategies produced by the system using a ranked learning Model, which is an example of a machine learning algorithm. Factors used to rank the tax strategies include (but are not limited to) projected tax savings, implementation likelyhood (eg. workflow feasibility), user preferences, any financial constraints on implementation, and estimated likelihood of acceptance.

4) Predictive Impact Modeling Engine - This module will predict the future financial implications of each tax planning strategy by evaluating projected income growth, potential changes in tax liability, and potential financial risks associated with each tax planner (e.g., using salary projection models and Monte Carlo simulation).

By integrating behavioural analysis and recommendation ranking along with predictive financial modeling, the proposed system will deliver customized and practical tax recommendations that are both attainable and likely to be implemented by taxpayers.

1.1.6 SDG Alignment

This research has a direct correlation with numerous United Nations Sustainable Development Goals (SDGs) related to facilitating sustainable economic growth, accountability of government actions, and provide guidance for responsible financial choices. Most notably, the proposed research supports the achievement of SDG 8 - Decent Work and Economic Growth, whose mission is to advance persistent economic progress, high-quality job creation, and enhancing the financial position of nations.

Efficient tax systems play an important role in ensuring the sustainable nature of economies, as successful completion of governmental objectives is predicated on tax revenue being the principal source of funding government provided services, building infrastructure, and developing national economic initiatives. Due to many people within society being unaware of tax laws/regulations or how to implement proper strategies for reducing income taxes, this research proposes to utilize an intelligent system that generates individualized tax recommendations and provides users with a forecast of how their financial decisions today will affect their future economic situation via future support on creating personal wealth.

The outcome of this research will ultimately promote healthy personal financial practices which, in turn, will enhance the overall economy. In addition, this research aligns with **SDG 16: Peace, Justice, and Strong Institutions**, which emphasizes transparency, accountability, and efficiency in governance systems.

Tax compliance and transparency are essential for maintaining trust between citizens and government institutions. When taxpayers have access to reliable decision-support systems that clearly explain tax-saving options and their financial consequences, they are more likely to comply with tax regulations and participate in the formal tax system.

The proposed system will improve the accessibility and transparency of the process of tax advice, which will assist in improving tax compliance and strengthening institutional accountability. Accordingly, this research contributes to economic sustainability as well as improved governance through better decision-support systems for taxpayers.

1.1.7 Research Cluster Alignment

This research falls within the realm of Artificial Intelligence (AI) and Data Science and will specifically be focusing on utilizing advanced machine learning methods for creating financial decision-support systems. As such, this research will combine the concepts of several different research domains including machine learning, recommender systems, financial technology (FinTech), predictive analytics, and intelligent decision-support systems.

The primary function of machine learning within the proposed system will be to analyze user financial profiles, predict user behaviour based upon their financial profiles, and recommend personalised recommendations to users. Techniques such as adoption probability prediction models and learning-to-rank algorithms will evaluate, and rank tax strategies based upon numerous decision factors including anticipated tax savings, financial feasibility, and user preferences.

Furthermore, predictive analytics techniques will be used to predict the long-term financial impact of various tax strategies. For instance, salary projection models and Monte Carlo simulations can be used by the system to identify potential future outcomes and estimate the impact that tax decisions may have on future tax liabilities, financial risk, and the growth of savings.

The integration of these technologies positions the proposed system within the broader FinTech research domain, where artificial intelligence is increasingly used to develop intelligent financial advisory platforms and automated decision-support tools. By combining behavioral modeling, recommendation systems, and predictive financial analysis, this research contributes to the development of intelligent financial advisory systems that support data-driven and personalized decision-making for individual taxpayers.

1.2 Literature Review

1.2.1 Review of Recent Scholarly Sources

Machine learning, recommendation systems and predictive analytics are increasingly being researched for their potential improvements to decision-making in finance and processing of tax-related issues. There have been a number of studies that have shown that machine learning methods can analyze large financial data sets and detect patterns which help to assist with the tax-related processes of compliance monitoring, detecting fraudulent activity and forecasting financial events.

Olabanji et al. [5] have discussed how machine learning technology can improve tax compliance and revenue collection due to the use of analyzing financial trends and identification of possible variances in tax data as well. In addition, Yang et al. [6] have demonstrated how predictive models can be used to detect potential tax compliance risks and improve efficiencies of tax systems. The above articles highlight the magnitude of the importance of the usage of data-driven technologies to an increasingly large degree by government agencies involved in the administration of taxation.

Across the board the adoption of recommendation systems have become commonplace to provide an avenue for individualised decision making to take place in the financial sector as well as in the commercial space. Of particular success has been the application of learning to rank algorithms to rank a number of alternatives to determine which one of the alternatives is most relevant to the user, based on the user's preferences and to determine the expected result of their selection. Scheltema [7] has explained how the use of ranking models within recommendation systems can affect the "sorting" of alternatives and which alternatives will be presented to the user. In a similar fashion to Scheltema [7], Wang [8] has examined how development of advanced ranking algorithms can improve the performance of recommendation systems through the use of information obtained from the user interaction data.

Recent research has additionally examined explainable and AI-based recommendation systems here. For example, Luo et al. [9] and Wu et al. [10] study techniques to rank recommendations in a way that increases the relevance and fairness of these recommendations to appoint users. Simultaneously, this emerging research demonstrates that providing transparent and explainable automated suggestions is an important consideration in using AI to develop recommendation systems.

Predictive modeling techniques are frequently applied in financial decision-support systems. In terms of forecasting based on simulations and how those simulations are applied, researchers have previously discussed how Monte Carlo simulation techniques can be utilized to evaluate financial risk as well as to estimate long-term financial outcomes in uncertain economic environments. Kumar et al. [13] illustrates that combining predictive analytics with simulation techniques provides a means of significantly improving financial projections and long-term financial decisions.

Despite these studies demonstrating the possible application of machine learning, recommendation systems, and predictive analytics to financial decision making, the available research demonstrates the limited applicability of these technologies to personalized tax advisory systems.

1.2.2 Existing Research Approaches

In general, prior research on the use of technology in tax compliance and providing assistance in financial decision-making is concerned with three approaches: 1) Tax Compliance Analysis, 2) Optimization-based Financial Advisory Systems, and 3) Recommendation Systems.

The first approach uses machine learning techniques to analyze financial or taxpayer-level data to detect potential compliance risks with tax laws. Tax-compliance analysis methods are designed to identify patterns of noncompliance (e.g., tax fraud, noncompliance alerts) in taxpayers (i.e., financial transaction-level) and taxpayer-level (i.e., taxpayer, taxpayer type) data, using methodologies such as statistical techniques or machine learning. Many of these tax-compliance analysis methods are designed for providing regulatory oversight of taxpayers rather than providing individualized decision support.

The second approach focuses on optimization-based financial advisory systems, which identify optimal financial strategies to implement in order to minimize tax liability or maximize financial return. However, optimization-based advisory systems generally assume that the taxpayer will implement the recommended strategies without taking into account the behavioral or other practical financial constraints associated with the implementation of a strategy.

The final approach, recommendation systems, leverage historical user preferences and behavior to provide individualized recommendations to the user. Learning to rank is a common algorithm used in recommendation systems for ranking multiple alternative options on the basis of multiple performance criteria. Recommendation systems are now commonly used in other areas of the economy such as e-commerce and online services, but are still very rarely utilized in the tax domain.

Although these research approaches contribute valuable insights, they often focus on individual aspects of financial decision-making, such as optimization, prediction, or recommendation, rather than integrating these capabilities into a unified tax advisory system.

1.2.3 Existing Products and Services

In practice, several tax advisory tools and financial software platforms have been developed to assist taxpayers with tax calculations, return filing, and understanding tax regulations. In Sri Lanka, the Inland Revenue Department (IRD)provides official guidelines, documentation, and online platforms that allow individuals to submit their annual income tax returns electronically and access information related to tax obligations and regulations [1], [2]. These systems are designed primarily to improve accessibility and simplify the tax filing process for taxpayers.

In addition to government-supported systems, various financial advisory platforms and tax calculation tools are available globally. These platforms typically allow users to enter financial information such as income, deductions, and expenses in order to estimate tax liabilities and identify possible tax deductions. Many of these systems use rule-based

calculations and predefined tax formulas to compute tax obligations based on the applicable tax regulations.

While these tools help taxpayers perform tax calculations and complete filing procedures more efficiently, they generally focus on tax compliance and reporting rather than strategic tax planning. Most existing products provide only basic guidance on deductions or tax rules and do not analyze individual financial behavior, preferences, or constraints when suggesting tax-saving options.

Furthermore, these systems rarely include advanced decision-support capabilities such as personalized tax strategy recommendations, behavioral analysis, or predictive modeling of future financial outcomes. Research shows that integrating machine learning and predictive analytics into financial systems can significantly improve decision-support capabilities and financial planning processes [5], [6].

As a result, taxpayers often receive general tax information rather than tailored recommendations that consider their financial situation and long-term goals. Consequently, most existing tax advisory products primarily support tax calculation and regulatory compliance, but they do not provide intelligent tools that assist individuals in selecting the most suitable tax strategies or evaluating the long-term financial impact of their tax decisions. This limitation highlights the need for more advanced tax advisory systems that integrate personalization, behavioral analysis, and predictive financial modeling [13].

1.2.4 Limitations of Existing Systems

Despite advancements in financial technology and data-driven decision-support systems, several limitations still exist in current tax advisory systems.

First, most existing systems rely on rule-based tax calculation methods, which apply predefined rules to determine tax liabilities but do not provide personalized strategy recommendations based on user behavior or financial preferences.

Second, many financial advisory systems focus primarily on maximizing tax savings without considering whether users are realistically able to adopt the recommended strategies. Factors such as financial flexibility, liquidity constraints, and risk tolerance are often ignored in existing systems.

Third, most tax advisory tools do not estimate the long-term financial impact of tax strategies. While predictive techniques such as Monte Carlo simulations can be used to evaluate financial risks and future outcomes [13], these methods are rarely integrated into tax advisory platforms.

Finally, existing systems rarely combine behavioral modeling, recommendation ranking, and predictive financial impact analysis into a single framework. Although learning-to-rank models have been widely used in modern recommendation systems to prioritize relevant options [7], [8], these techniques are not commonly applied to tax strategy recommendation systems.

These limitations highlight the need for intelligent systems that can generate personalized, behavior-aware, and future-oriented tax recommendations, which motivates the development of the proposed Personalized Recommendation and Predictive Impact Modeling Engine.

1.3 Research Gap

1.3.1 Gap in Personalized Tax Advisory Systems

Most of the existing tax advisory systems have concentrated on rule-based tax calculation and compliance, and there is little focus on decision support for taxpayers. The existing government tax systems and digital tax-related technologies have concentrated on assisting taxpayers in calculating taxes and filing tax returns, as per regulations [1], [2]. Although these technologies have helped taxpayers by making tax-related services more accessible and simplifying tax-related processes, there is no focus on intelligent decision support, depending on the financial profile and preferences of taxpayers.

There is a research gap identified for developing personalized tax advisory systems that can assist taxpayers in generating tax savings strategies using user-specific financial information

1.3.2 Gap in Behavioral Adoption Prediction

However, it is seen that the existing financial advisory systems do not consider the behavioral aspects of the user in adopting the recommended strategies. Financial decisions are affected by various factors such as risk tolerance, liquidity constraints, financial flexibility, and personal financial habits.

Although various machine learning models have been proposed for the analysis of financial data and tax compliance patterns [5], [6], there is a lack of consideration of behavioral modeling in predicting the probability of adopting the recommended strategies in the existing tax advisory systems.

This is a significant research gap in developing a tax advisory system that incorporates a model for predicting the probability of adopting the recommended strategies.

1.3.3 Gap in Strategy Ranking and Decision Prioritization

Another weakness of existing systems is that they do not offer adequate tools for evaluating and prioritizing multiple tax strategies. Some of these tools present lists of potential tax savings, but they do not offer a prioritized list of tax savings based on their suitability for each user.

In contemporary recommendation systems, learning-to-rank models are popular for prioritizing options according to various factors such as benefits, user preferences, and feasibility [7], [8]. However, they are not commonly utilized in tax recommendation systems.

This indicates a research gap in utilizing learning-to-rank models for prioritizing tax strategies according to tax savings, feasibility, and user preferences.

1.3.4 Gap in Long-Term Financial Impact Prediction

Most current tax advisory tools focus mainly on short-term tax calculations, such as estimating annual tax liabilities or identifying immediate deductions. However, tax decisions often have long-term financial consequences that influence future tax liabilities, savings, and financial risks.

Predictive modeling techniques such as Monte Carlo simulation are widely used in financial forecasting to evaluate long-term financial outcomes under uncertain conditions [13]. Despite their effectiveness in financial planning, these techniques are rarely integrated into tax advisory systems.

Therefore, there is a research gap in developing systems that can predict the long-term financial impact of tax strategies and provide forward-looking decision support.

1.3.5 Integrated Gap in Intelligent Tax Decision-Support Systems

While various researchers have studied and explored different machine learning applications in taxation, recommendation systems, and finance, they are usually considered independently and not in an integrated manner.
Research has been conducted on various aspects of machine learning and its applications in taxation, recommendation systems, and finance, including:

- Tax compliance analysis
- Financial forecasting
- Recommendation algorithms

However, not much research has been conducted on an integrated system of personalized recommendation generation, behavioral adoption prediction, strategy ranking, and predictive financial impact modeling, which is considered by the proposed Personalized Recommendation and Predictive Impact Modeling Engine for providing personalized tax recommendations for individual taxpayers.

1.3.6 Justification of Novelty and Creativity

The proposed research introduces a novel approach by integrating behavior-aware recommendation systems, predictive financial modeling, and machine learning-based ranking techniques into a unified tax advisory framework. Unlike traditional tax advisory systems that focus solely on tax calculations, the proposed system aims to provide personalized, practical, and future-oriented tax strategy recommendations.

One key novelty of this research is the integration of an adoption probability model, which predicts the likelihood that a taxpayer will follow a recommended strategy. This approach

addresses an important limitation in existing systems, where technically optimal strategies may fail in practice because they do not align with user behavior or financial constraints.

Another innovative aspect of the proposed system is the use of learning-to-rank algorithms to evaluate and prioritize tax strategies according to multiple criteria, including expected tax savings, feasibility, adoption likelihood, and financial risk. Learning-to-rank techniques have been widely used in recommendation systems in domains such as e-commerce and digital services, but their application in tax advisory systems remains limited [7], [8].

The research also introduces predictive impact modeling, which estimates the long-term financial consequences of adopting different tax strategies. By incorporating salary projection models and Monte Carlo simulations, the system can evaluate multiple future scenarios and provide users with insights into potential financial outcomes over time.

By combining these techniques into a single intelligent framework, the proposed system offers a more comprehensive and practical approach to tax decision-support, representing a significant advancement compared to existing tax advisory solutions.

1.3.7 Evidence-Based Argument

The proposed research aims to provide a novel approach in the field of tax advisory systems by incorporating behavior-aware recommendation systems, predictive financial modeling, and ranking mechanisms using machine learning algorithms. Compared to traditional tax advisory systems, the proposed approach is beneficial in the sense that it provides tax strategy recommendations in a personal, useful, and forward-looking manner.

One of the main contributions of the proposed research is the development of a model that predicts the probability of adopting a particular tax strategy. This is a significant improvement over traditional tax advisory systems, in which the recommended strategies may not be effective in practice due to their incompatibility with user behavior.

Another significant contribution of the proposed research is the development of a tax strategy evaluation and ranking mechanism using learning-to-rank algorithms. These algorithms are useful in evaluating tax strategies considering various parameters such as tax reduction, feasibility, and probability of adoption. Learning-to-rank algorithms have already been implemented in recommendation systems for various applications, such as e-commerce and online services. However, their use in tax advisory systems is still in its infancy [7], [8].

The research also introduces predictive impact modeling, which estimates the long-term financial consequences of adopting different tax strategies. By incorporating salary projection models and Monte Carlo simulations, the system can evaluate multiple future scenarios and provide users with insights into potential financial outcomes over time.

Additionally, this research introduces predictive impact modeling, which is used for predicting the financial impact of adopting various tax strategies in the future. By using salary projection models and Monte Carlo simulations, it is possible for this system to analyze various future scenarios and give users insights into potential financial outcomes.

With this integration of various techniques into a single intelligent system, it is evident that this proposed system provides a more comprehensive and practical approach for tax decision-support systems, marking a significant improvement from existing tax advisory systems

1.3.8 System comparison (Existing Systems vs Proposed System)

Feature	Tax Portals (IRD e-Services / RAMIS) [1][3][4]	AI-based Tax Compliance Research [5][6][9]	Digital Tax Administration Frameworks (OECD Tax Administration 3.0) [8]	Tax Planning & Advisory AI Research [7][10]	Proposed System (Personalized Recommendation & Predictive Impact Engine)
Online tax filing support	Yes	No	Partial	Partial	Yes
Tax compliance monitoring	Partial	Yes	Yes	Partial	Yes
Automated transaction interpretation	No	Limited	Limited	No	Yes
Personalized tax strategy recommendations	No	No	No	Partial	Yes
Behavioral adoption modeling	No	No	No	No	Yes
Predictive tax impact modeling	No	Limited	No	Limited	Yes
Multi-objective optimization (Savings × Risk × Adoption)	No	No	No	No	Yes
Future financial projection capability	No	Limited	No	Limited	Yes
Explainable recommendation reasoning	No	Partial	Partial	Partial	Yes
Decision-support for taxpayers	Limited	Limited	Limited	Partial	Yes
Sri Lankan tax regulation integration	Yes	No	No	No	Yes

Table A : System Comparison

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to create an intelligent Personalized Recommendation and Predictive Impact Modeling Engine that helps taxpayers make informed decisions in income tax optimization. This proposed system aims to analyze personal financial information, possible tax strategies, and personal behavioral characteristics to provide personalized tax recommendations that maximize expected tax savings for taxpayers in Sri Lanka.

To achieve this objective, the research aims to create a decision-support framework for evaluating various tax strategies and ranking them according to various significant criteria such as expected tax savings, financial feasibility, financial risks, and the likelihood of a user adopting a particular tax strategy. By incorporating personal behavioral characteristics and advanced recommendation technologies such as learning to rank algorithms, the proposed decision-support framework aims to provide tax recommendations that are theoretically optimal and practically feasible for individual taxpayers.

In addition to that, a predictive modeling system of financial impact will be included in the proposed system to assist taxpayers in understanding the possible impact of their current tax decisions on their future financial well-being. To do that, the proposed system will utilize various modeling techniques, including salary projection modeling and simulation forecasting techniques, to determine possible tax liabilities that might arise in the future under various scenarios of adopting tax strategies.

Through the achievement of that goal, it is hoped that the proposed intelligent tax advisory system can assist taxpayers in understanding tax savings opportunities in a clear manner. This can lead to improved voluntary compliance and decision-making in relation to tax issues.

2.2 Specific Objectives

2.2.1 Develop a Personalized Tax Recommendation Ranking Model

The first objective is to create and implement a learning-to-rank-based recommendation system that is able to evaluate various tax savings strategies and rank them according to their relevance and effectiveness for a particular taxpayer.

This will be achieved by creating a ranking model that will analyze various evaluation factors, such as expected savings, financial feasibility, user preferences, and tax strategy applicability.

The ranking model will be created using simulated user profiles and financial scenarios, reflecting various user profiles and financial behaviors, to ensure that it is able to identify the most suitable tax strategies for particular taxpayers.

2.2.2 Build an Adoption Probability Prediction Model for Tax Strategies

The second objective is to build a machine learning-based model for predicting adoption, which will predict the likelihood of a taxpayer adopting a recommended tax strategy.

This model will assess various parameters, such as income level, flexibility, behavioral factors, and risk tolerance, and then predict whether a user would adopt a recommended strategy or not. The adoption probability score will also be factored in while determining the rankings of recommended strategies so that they are feasible and doable by users.

2.2.3 Implement a Predictive Financial Impact Estimation Module

Another objective is to design a predictive financial modeling component that estimates the potential long-term financial impact of adopting different tax strategies.

Another aim would be to include a predictive financial modeling tool that would be able to forecast the potential financial implications of adopting different tax strategies.
This module would include a salary projection model that would be able to forecast future salary increases and potential future tax liabilities under different financial scenarios.
In addition, simulation techniques such as Monte Carlo simulation would be utilized in analyzing different potential financial outcomes and forecasting potential future benefits of different tax strategies.

2.2.4 Develop a Multi-Objective Scoring Framework for Tax Strategy Evaluation

This objective is concerned with creating a multi-objective evaluation system that is able to evaluate tax strategies based on various factors for decision-making.

This proposed scoring function will enable evaluation of tax strategies based on various factors such as tax savings, probability of adoption, financial feasibility, and financial risks. By using this scoring function, it will be possible to develop optimized and personalized tax recommendations based on various factors for decision-making and not limited to tax savings only.

personalized tax recommendations that balance multiple decision objectives rather than focusing solely on tax savings.

2.2.5 Design a Personalized Decision-Support Dashboard for Taxpayers

Another objective is to create a user-friendly decision support system that enables taxpayers to easily interact with and comprehend the proposed decision support system.

The proposed decision support system will enable users to access a list of tax strategies in ranked order along with descriptions, predicted financial results, and confidence levels that are generated by the system. It will also enable users to evaluate various financial scenarios to determine how they might impact their tax liabilities in the future.

2.2.6 Evaluate the Effectiveness of the Recommendation and Predictive Modeling System

Finally, the ultimate objective is to assess the performance of the proposed system for recommendation and predictive modeling.

To assess the performance of the proposed system for recommendation and predictive modeling, the system will be tested using simulated financial data sets. Real taxpayer data is not available due to privacy restrictions.

Various metrics are used to evaluate the performance of the system. These metrics include the accuracy of the predictions made by the system, the relevance of the recommendations provided by the system, the performance of the system in ranking the results, and the improvement of the estimated tax savings using the system compared to the conventional rule-based tax advisory systems.

The performance of the system is essential to assess the success of the proposed system in improving the personalized tax recommendations.

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3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, which is widely used in information systems research for developing innovative technological artifacts that address real-world problems. Design Science Research focuses on designing, building, and evaluating systems or models that provide practical solutions while contributing new knowledge to the research domain.

In this study, the primary artifact is a Personalized Recommendation and Predictive Impact Modeling Engine designed to assist taxpayers in making informed income tax decisions. The design science approach is suitable for this research because the objective is to develop an intelligent system that integrates machine learning algorithms, predictive modeling techniques, and decision-support mechanisms to improve personalized tax advisory services.

The research follows an **iterative design cycle**, consisting of the following stages:

1 **Problem Identification**: Analyze the limitations of existing tax advisory systems and identify the need for personalized tax recommendation and predictive decision-support tools.

2 **Design and Development**: Design and implement machine learning models, recommendation algorithms, and predictive modeling components that generate personalized tax strategies.

3 **Demonstration** : Implement the proposed system architecture and demonstrate its functionality using simulated financial datasets and user scenarios.

4 **Evaluation** :Evaluate system performance using quantitative metrics such as prediction accuracy, recommendation relevance, and ranking effectiveness.

5 **Communication**: Document and present the research findings, methodology, and system contributions through academic reports and presentations.

This iterative research process allows continuous refinement and improvement of the system until the desired performance and usability levels are achieved.

3.2 System Architecture Overview

The proposed system consists of four main modules that interact to provide personalized tax decision-support for taxpayers.

1. User Financial Profile Module

This module collects and stores user financial information, including:

- Income sources
- Salary history
- Tax deductions
- Financial preferences
- Risk tolerance

The collected data is used to build a structured financial profile representing the user's financial behavior and decision-making characteristics.

2. Tax Strategy Recommendation Engine

This module generates multiple potential tax strategies based on applicable tax regulations. The generated strategies are evaluated and ranked using machine learning algorithms to identify the most suitable options for each user.

3. Predictive Impact Modeling Module

This module estimates the future financial consequences of adopting different tax strategies. It uses financial projection models and simulation techniques to forecast future tax liabilities and financial outcomes.

4. Decision Support Dashboard

The dashboard presents ranked tax strategies along with explanations, predicted tax savings, and confidence scores. It allows users to explore different financial scenarios and understand the potential impact of their tax decisions.

Together, these modules form a personalized tax decision-support system capable of analyzing financial scenarios and generating actionable recommendations.

3.3 Technical Methods and Algorithms

The proposed system integrates several machine learning and data analysis techniques to generate personalized tax recommendations.

3.3.1 Learning-to-Rank Recommendation Model

A **learning-to-rank algorithm** will be implemented to prioritize tax strategies according to their relevance and expected benefit for individual users.

The ranking model will evaluate tax strategies using several factors, including:

- Expected tax savings
- Financial risk
- Adoption likelihood
- Future financial impact

Learning-to-rank models are widely used in recommendation systems to rank alternatives based on multiple decision factors and contextual information.

3.3.2 Adoption Probability Prediction Model

A **supervised machine learning classifier** will be developed to estimate the probability that a user will adopt a recommended tax strategy.

Input features may include:

- Income level
- Financial flexibility
- Risk tolerance
- Historical financial behavior patterns

The predicted **adoption probability score** will be integrated into the recommendation ranking process to ensure that recommended strategies are both beneficial and realistic for users to follow.

3.3.3 Predictive Financial Impact Modeling

To estimate long-term financial outcomes, the system will implement **financial projection models combined with simulation techniques**.

The predictive model will:

- Estimate future income growth
- Calculate projected tax liabilities under different financial scenarios
- Forecast long-term tax savings for each recommended strategy

Monte Carlo simulation will be used to analyze uncertainty in financial variables such as salary growth and investment returns, enabling the system to evaluate multiple possible financial outcomes.

3.3.4 Multi-Objective Scoring Function

A **multi-objective scoring function** will be designed to evaluate tax strategies based on several criteria simultaneously.

The scoring function can be represented as:

Score = f(Tax Savings, Adoption Probability, Risk Level)

This evaluation approach ensures that the recommended strategies maximize financial benefits while also considering practical feasibility and user behavior.

3.4 Tools and Platforms

The system will be developed using the following technologies:

Component	Tools / Technologies
Programming Language	Python
Machine Learning Libraries	Scikit-learn, TensorFlow, PyTorch
Data Processing	Pandas, NumPy
Backend API	FastAPI
Database	MongoDB / PostgreSQL
Visualization Dashboard	React / Streamlit
Experiment Environment	Jupyter Notebook

Table B: Tools and Platforms

These tools provide a flexible environment for implementing machine learning models, managing data processing pipelines, and developing interactive decision-support dashboards.

3.5 Data Collection and Ethical Considerations

Due to privacy and regulatory restrictions, access to real taxpayer financial data is limited. Therefore, this research will primarily use **synthetic financial datasets and simulated user profiles** for experimentation and system evaluation.

The dataset will include:

- Simulated financial transaction records
- Synthetic income distributions based on tax regulations
- Simulated behavioral responses representing taxpayer decision patterns

Ethical Considerations

The research will follow strict ethical guidelines to ensure responsible data usage:

- No personally identifiable financial data will be used
- All datasets will be synthetic or anonymized
- Data privacy and confidentiality will be maintained throughout the research process

These measures ensure compliance with data protection regulations while enabling realistic system evaluation.

3.6 Validation Strategy and Performance Metrics

The proposed system will be evaluated using quantitative performance metrics and scenario-based experiments.

Metric	Purpose
Recommendation Accuracy	Measures correctness of recommended strategies
Prediction Accuracy	Evaluates adoption probability model performance
Expected Tax Savings	Compares tax savings with baseline rule-based strategies
Ranking Quality (NDCG / MAP)	Measures effectiveness of recommendation ranking
Simulation Reliability	Validates predictive financial impact models

Table C : Quantitative Performance Metrics

4. HIGH-LEVEL SYSTEM ARCHITECTURE

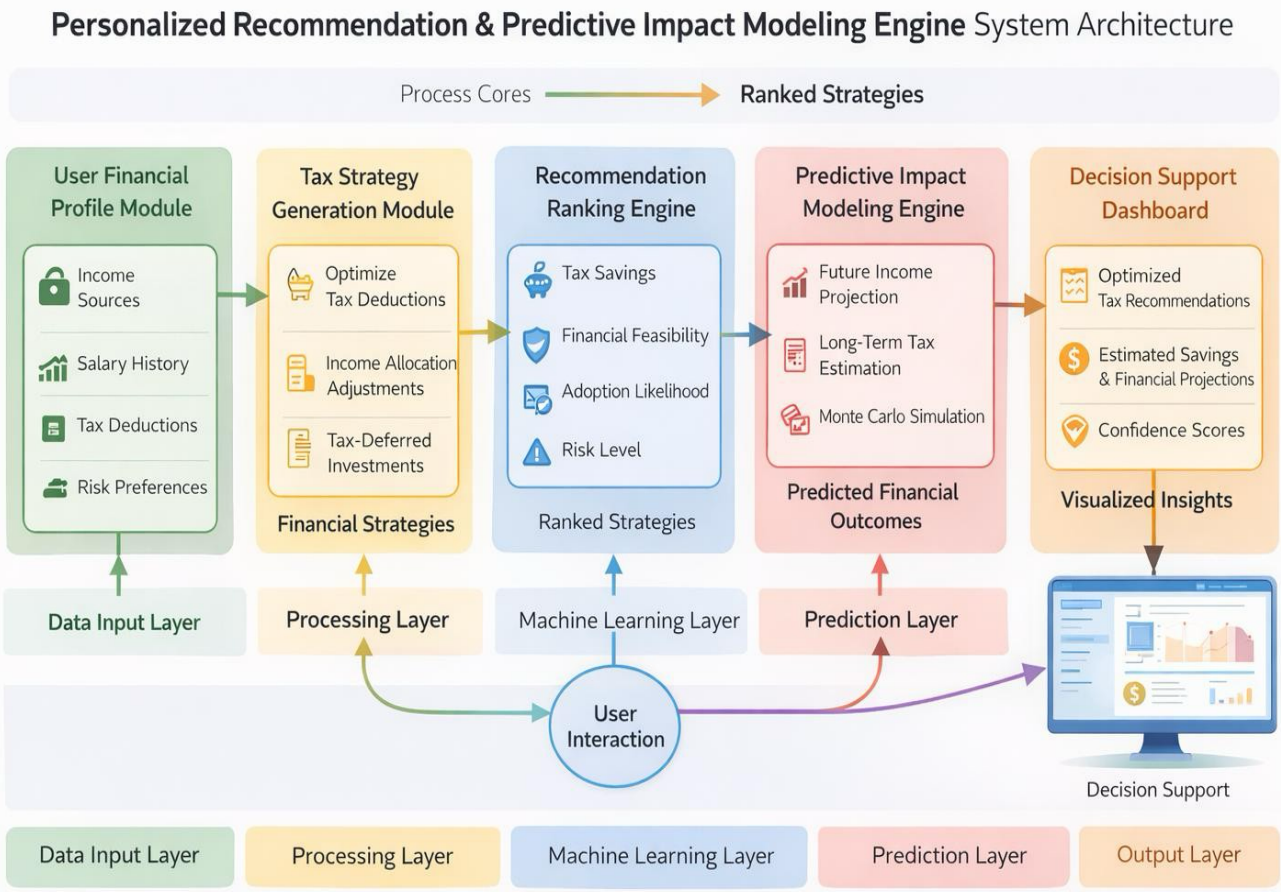


Figure 1 - Personalized Tax Recommendation & Impact Modeling System Architecture

4.1 Overview of the Proposed Solution

The proposed system introduces a Personalized Recommendation and Predictive Impact Modeling Engine designed to support income tax decision-making for individual taxpayers. The system analyzes financial data, evaluates potential tax strategies, and generates personalized recommendations using machine learning and predictive modeling techniques.

The architecture integrates financial information processing, recommendation algorithms, and predictive simulation models to deliver transparent and explainable tax decision support. The system evaluates multiple tax strategies and ranks them using a learning-to-rank recommendation algorithm, while also estimating long-term financial outcomes using predictive impact modeling.

Modern recommendation architectures typically combine data processing layers, machine learning models, and real-time recommendation services to deliver personalized insights efficiently. Following this design principle, the proposed architecture consists of five main components:

- User Financial Profile Module
- Tax Strategy Generation Module
- Recommendation Ranking Engine
- Predictive Impact Modeling Engine *(Primary Research Component)*
- Decision Support Dashboard

Together, these components form an intelligent decision-support platform that helps taxpayers analyze financial scenarios, evaluate tax strategies, and optimize their tax planning decisions.

4.2 System Components

The system architecture consists of five interconnected modules that process financial information and generate personalized tax recommendations.

4.2.1 User Financial Profile Module

This module collects, stores, and manages financial data related to individual taxpayers.

Input data includes:

- Income sources
- Salary history
- Tax deductions
- Financial behavior indicators
- Risk tolerance preferences

The module organizes this information into structured datasets that represent the user's financial profile. These datasets are then used as inputs for the recommendation and predictive modeling modules.

4.2.2 Tax Strategy Generation Module

This module generates potential tax planning strategies that may be applicable to the user's financial situation.

Strategies are generated based on:

- Sri Lankan tax regulations
- Available deduction opportunities
- Financial eligibility rules

Examples of generated strategies include:

- Claiming applicable tax deductions
- Adjusting income allocation methods
- Utilizing available tax exemptions and benefits

These candidate strategies are forwarded to the recommendation engine for evaluation and ranking.

4.2.3 Recommendation Ranking Engine

The recommendation engine evaluates and prioritizes generated tax strategies using machine learning-based ranking techniques.

A learning-to-rank algorithm is used to order strategies based on multiple criteria, including:

- Expected tax savings
- Financial feasibility
- User preferences
- Adoption probability

Learning-to-rank models are widely used in recommendation systems to determine the most relevant options for users by ordering alternatives according to predicted relevance.

The output of this module is a ranked list of personalized tax strategies tailored to each taxpayer.

4.2.4 Predictive Impact Modeling Engine

The Predictive Impact Modeling Engine is the core research component of this study. This module estimates the future financial consequences of adopting different tax strategies.

The engine performs the following analyses:

- Income projection modeling
- Future tax liability estimation
- Monte Carlo simulation of financial scenarios

The outputs generated by this module include:

- Estimated long-term tax savings
- Financial risk assessment
- Future financial projections

This predictive layer enables taxpayers to understand how their current tax decisions may influence their financial outcomes in future tax years.

4.2.5 Decision Support Dashboard

The Decision Support Dashboard provides a user-friendly interface that allows taxpayers to visualize and interact with the system's recommendations.

The dashboard presents:

- Ranked tax strategies
- Predicted financial outcomes
- Estimated tax savings
- Confidence scores for model predictions

Users can explore alternative financial scenarios and compare potential tax planning strategies, enabling informed and transparent decision-making.

4.3 Component Interactions

The system components interact in a sequential workflow to generate personalized tax recommendations.

The interaction process occurs as follows:

1 User financial data is collected and processed by the User Financial Profile Module.

2 The Tax Strategy Generation Module creates possible tax planning strategies based on tax regulations and financial eligibility rules.

3 The Recommendation Ranking Engine evaluates and ranks these strategies according to multiple criteria.

4 The Predictive Impact Modeling Engine estimates long-term financial outcomes associated with each strategy.

5 The Decision Support Dashboard presents the ranked strategies and predictive insights to the user.

This interaction forms a decision-support loop where user feedback and behavioral data can be used to continuously refine future recommendations.

4.4 Data Flow Explanation

The system processes financial data through a multi-layer pipeline to generate personalized tax recommendations.

1. Data Input Layer

This layer collects financial information including:

- Financial records
- Income details
- Taxpayer profiles

2. Processing Layer

The system performs:

- Feature extraction
- Financial scenario modeling

3. Machine Learning Layer

Machine learning models perform:

- Adoption probability prediction
- Recommendation ranking

4. Prediction Layer

The predictive modeling module estimates:

- Potential tax savings
- Long-term financial projections

5. Output Layer

The system produces:

- Personalized tax strategy recommendations
- Visualized financial impact insights

This pipeline transforms raw financial data into actionable decision-support insights.

4.5 Scalability and Security Considerations

1. Scalability

The system is designed to support large numbers of users and financial datasets through a scalable cloud-based architecture.

Key scalability strategies include:

- Distributed machine learning model training
- Scalable database storage
- Microservice-based system architecture
- API-based system integration

These approaches enable the system to efficiently handle increasing volumes of financial data and user interactions.

2. Security

Because the system processes sensitive financial data, strong security mechanisms are required to protect taxpayer information.

Security measures include:

- Encryption of financial records
- Secure authentication mechanisms
- Role-based access control
- Privacy-preserving data storage

These mechanisms ensure that taxpayer data remains confidential, secure, and protected from unauthorized access.

5. USER REQUIREMENTS

5.1 Overview of User Requirements

The requirements of users of this component are based on the practical needs of individuals who need to take appropriate decisions regarding their income tax planning and management. The proposed system intends to assist users in making appropriate decisions regarding their tax management by analyzing various aspects of their finances and determining appropriate tax strategies to ensure optimal results in compliance with tax laws in Sri Lanka.

The system intends to transform various pieces of data related to finances and tax into tax decision support using machine learning and predictive modeling technologies. The proposed component provides users with ranked tax strategies, predictive analysis of financial impact, and tax decision support instead of tax calculation results. This way, users can get a clear understanding of how tax decisions affect their financial life in the future.

Considering the proposed system's requirements, users of this component are individuals who are either taxpayers or tax consultants, as well as various modules of the project that use tax decision support results.

The requirements are defined to ensure that the system addresses real user needs rather than functioning purely as a technical research prototype. Therefore, the requirements emphasize not only functional capabilities such as tax strategy recommendation and predictive modeling, but also important quality attributes such as usability, transparency, privacy, reliability, and performance.

Table D – Primary Users of the Component

User Type	**Role**
Individual taxpayer	Receives personalized tax strategy recommendations and financial impact insights
Tax advisor / financial consultant	Reviews recommended strategies and supports tax planning decisions
Other system components	Consume structured recommendation outputs for integration with other modules

5.2 Functional Requirements

The functional requirements describe the key capabilities that the system must provide in order to generate personalized tax recommendations and predictive financial insights.

ID	Functional Requirement	Description
FR1	User Financial Data Input	The system shall allow users to input financial information such as income sources, salary history, tax deductions, and financial preferences.
FR2	Financial Profile Creation	The system shall organize user financial information into a structured financial profile for analysis and recommendation generation.
FR3	Tax Strategy Generation	The system shall generate potential tax planning strategies based on applicable Sri Lankan tax regulations and deduction opportunities.
FR4	Strategy Feasibility Analysis	The system shall evaluate whether generated tax strategies are financially feasible based on the user's financial profile.
FR5	Recommendation Ranking	The system shall rank candidate tax strategies using machine learning techniques based on factors such as expected tax savings and adoption probability.
FR6	Adoption Probability Prediction	The system shall estimate the likelihood that a user will adopt a recommended tax strategy using behavioral prediction models.
FR7	Predictive Financial Impact Estimation	The system shall estimate the long-term financial impact of adopting different tax strategies, including projected tax savings and financial risks.
FR8	Scenario Simulation	The system shall simulate alternative financial scenarios to estimate future tax outcomes using predictive modeling techniques.
FR9	Multi-objective Strategy Evaluation	The system shall evaluate tax strategies using a scoring model that considers tax savings, risk level, and adoption likelihood.
FR10	Recommendation Explanation	The system shall provide explainable outputs that describe why a specific tax strategy was recommended.
FR11	Decision Support Visualization	The system shall present ranked tax strategies and financial projections through an interactive dashboard interface.
FR12	Structured Output Generation	The system shall generate structured recommendation outputs that can be consumed by other modules of the overall project architecture.

Table E: Functional Requirements

5.3 Non-Functional Requirements

The non-functional requirements describe the quality attributes and system properties that ensure the reliability, usability, and security of the system. These requirements ensure that the system is not only technically functional but also practically usable for taxpayers and financial advisors.

ID	Non-Functional Requirement	Description
NFR1	Performance	The system should generate personalized tax recommendations within an acceptable response time suitable for prototype decision-support applications.
NFR2	Accuracy and Reliability	The recommendation ranking model and prediction models should achieve acceptable accuracy based on evaluation metrics defined in the methodology section.
NFR3	Explainability	The system should provide clear explanations for recommended tax strategies to ensure transparency and user understanding.
NFR4	Privacy and Data Security	The system must protect sensitive financial information using encryption, secure storage, and controlled access mechanisms.
NFR5	Scalability	The architecture should support large numbers of users and financial datasets without significant performance degradation.
NFR6	Maintainability	The system should be modular so that recommendation algorithms, predictive models, and data processing modules can be updated independently.
NFR7	Usability	The dashboard interface should present recommendations and financial insights in a clear and understandable manner for non-expert users.

NFR8	Robustness	The system should remain functional even when financial inputs are incomplete, inconsistent, or noisy.
NFR9	Interoperability	The system should provide outputs that can be integrated with other components in the overall project architecture.
NFR10	Auditability	The system should retain intermediate reasoning and scoring information to support validation, debugging, and result explanation.

Table F : Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Commercialization Overview

The proposed component, namely Personalized Recommendation and Predictive Impact Modeling Engine, has high commercialization potential as a component of an artificial intelligence-based tax advisory and financial decision-support module. The commercial value of this component is in its potential to transform complex financial information into personalized tax strategies and predictive financial insights for individual taxpayers.

The proposed component is differentiated from existing tax portals and tax calculators, which primarily facilitate tax filing and tax calculations, respectively. The proposed component provides intelligent recommendation capabilities through its integration of machine learning models, behavioral prediction, and predictive financial simulation. It analyzes a user's financial profile, evaluates various tax planning strategies, and provides ranked recommendations based on tax savings, financial feasibility, and predicted adoption likelihood.

From a commercialization perspective, this component can be commercialized as a decision-support intelligence engine and not as a tax filing platform. The proposed component can be commercialized in various real-world scenarios as follows:

- A backend recommendation API for digital tax advisory platforms
- An AI-powered tax optimization module for financial management applications
- An enterprise decision-support tool for accounting and tax consulting firms
- A feature layer integrated into fintech platforms that provide tax planning services

This approach is commercially attractive because the component addresses a major challenge faced by taxpayers, understanding and selecting practical tax-saving strategies based on their financial situation.

6.2 Value Proposition

The main value proposition of the proposed component is that it enables personalized and data-driven tax planning by transforming financial information into optimized tax strategies and predictive financial insights.

The system offers capabilities that go beyond traditional tax calculators by providing behavior-aware recommendations and long-term financial impact analysis. Instead of simply calculating tax liabilities, the system evaluates multiple tax planning strategies and helps users understand the consequences of their financial decisions.

The component creates value in several ways:

- It helps taxpayers identify tax-saving opportunities that match their financial profile.
- It reduces the complexity of tax planning by providing ranked recommendations instead of generic advice.
- It enables predictive financial analysis, helping users understand how tax decisions may influence future financial outcomes.

• It supports better financial planning and tax compliance through transparent and explainable recommendations.
• It can be integrated into existing financial systems to provide AI-powered tax decision-support capabilities.

This value proposition is commercially important because it improves both decision quality and financial planning efficiency for taxpayers and financial professionals.

6.3 Target Market

The commercialization strategy for this component can target both business-oriented markets and individual users.

6.3.1 Primary Target Market

The primary target market consists of organizations and professional users that provide tax advisory or financial planning services.

These include:

- **Tax advisory firms and tax consultants**
  Professionals who assist individuals and businesses in tax planning and tax optimization.
- **Accounting and financial consulting firms**
  Organizations that manage financial records and provide financial planning advice.
- **Fintech and tax-tech platforms**
  Companies developing digital tax assistance tools that require intelligent recommendation systems.
- **Personal finance management platforms**
  Software vendors providing financial planning tools that could integrate personalized tax advisory features.

These markets are ideal for early commercialization because they already manage financial data and require advanced tax decision-support tools.

6.3.2 Secondary Target Market

The secondary market includes individual users and organizations that could benefit from automated tax advisory services.

These include:

• Individual taxpayers with complex financial situations
• Freelancers or professionals with multiple income sources
• Small business owners managing personal and business income
• Financial planning service providers

Although these users may benefit from the system, commercialization efforts may initially focus on B2B markets, where integration with financial platforms can scale adoption more effectively.

6.4 Potential Customers and Users

In the early commercialization phase, the paying customers and the end users may differ.

Likely Paying Customers

• Accounting firms
• Tax consulting companies
• Financial advisory service providers
• Fintech companies developing tax-support applications
• Personal finance management platforms

Likely End Users

• Individual taxpayers
• Tax advisors
• Financial consultants
• Accountants
• Advanced personal finance users

Understanding this distinction is important because organizations may purchase the system while individual users interact with the recommendation outputs.

6.5 Unique Selling Proposition (USP)

The unique selling proposition of the proposed component is its ability to provide personalized, predictive, and explainable tax strategy recommendations rather than simple tax calculations.

Key differentiating features include:

• Personalized tax strategy recommendations based on individual financial profiles
• Behavior-aware recommendation models that predict whether users are likely to adopt a strategy
• Learning-to-rank recommendation algorithms that prioritize the most suitable tax strategies
• Predictive financial impact analysis using income projections and simulation models
• Multi-objective optimization considering tax savings, risk level, and adoption likelihood
• Explainable recommendation reasoning that helps users understand why a strategy is suggested
• Integration capability with financial platforms and fintech systems

From a commercial perspective, the system is not simply a tax calculator, it is an AI-driven tax planning and financial decision-support engine.

6.6 Revenue Model

Several revenue models can be used to commercialize the proposed system.

1. Subscription-Based Licensing (B2B)

Accounting firms, tax consultants, and financial platforms can subscribe to the system on a monthly or annual basis to access the recommendation engine and predictive financial analysis features.

2. API-Based Usage Pricing

Organizations can integrate the system through APIs and pay based on usage, such as:

• number of recommendation requests
• number of users
• number of financial analyses performed

3. Enterprise Integration Services

Custom integration, local deployment, and system customization for large financial platforms can generate one-time integration and deployment fees.

4. Premium Financial Analytics Features

Advanced capabilities such as:
• financial scenario simulation
• tax planning reports
• predictive financial dashboards
can be offered as premium features for professional users.

7. BUDGET AND JUSTIFICATION

The proposed budget covers the resources required to design, develop, train, evaluate, and demonstrate the Personalized Recommendation and Predictive Impact Modeling Engine within the research timeline. Since the proposed system is primarily a software-based machine learning solution, the budget focuses on computing infrastructure, dataset preparation, experimentation resources, and prototype development rather than large hardware investments.

The project requires computing resources for machine learning model training, recommendation algorithm development, and predictive financial simulations, as well as storage for datasets and experimental outputs. Additionally, resources are allocated for synthetic dataset preparation, prototype testing, and research support activities.
Each cost item listed in the budget directly contributes to the successful implementation, evaluation, and demonstration of the proposed system.

7.1 Estimated Budget

Table G – Estimated Budget for the Component

Resource Type	Purpose	Estimated Amount (LKR)
Cloud / GPU compute credits	Machine learning model training, experimentation, and predictive simulation processing	50,000
Data storage and backup	Storage of financial datasets, trained models, experiment outputs, and system logs	10,000
Internet and online research support	Access to research papers, datasets, cloud services, and development tools	12,000
Software / API / development tools	Experiment tracking tools, ML libraries, collaboration tools, and API services	8,000
Synthetic dataset generation and preparation	Creating simulated financial profiles, tax scenarios, and behavioral datasets for model training	15,000
Testing and prototype demonstration expenses	Dashboard development, visualization tools, demo preparation, and evaluation reports	10,000
Contingency allocation	Unexpected research expenses such as additional compute usage or data storage	10,000

Total Estimated Budget: 115,000

7.2 Budget Justification

Cloud / GPU Compute Credits

- Machine learning models used for recommendation ranking, adoption prediction, and predictive financial impact analysis require significant computational resources during model training and experimentation. Cloud computing platforms provide access to GPU-enabled environments that can accelerate model training and simulation processes. Using cloud services is more cost-effective than purchasing dedicated high-performance hardware.

Data Storage and Backup

- The system will generate several datasets including synthetic financial datasets, tax strategy simulations, model outputs, and experimental results. Secure storage and backup facilities are required to ensure that experimental data, trained models, and evaluation outputs are safely stored and protected from data loss.

Internet and Online Research Support

- This allocation supports continuous access to online research papers, machine learning libraries, datasets, and development tools required for implementing and evaluating the system. Reliable internet access is necessary for downloading research resources, accessing cloud services, and collaborating during the development process.

Software / API / Development Tools

- Although most machine learning frameworks such as Python, Scikit-learn, TensorFlow, and PyTorch are open-source, certain productivity tools, API services, or development utilities may require small subscription costs. This budget also supports software tools for experiment tracking, visualization, and prototype integration.

Synthetic Dataset Generation and Preparation

- Because real taxpayer financial data is confidential and restricted, the system will rely on synthetic financial datasets and simulated user profiles for experimentation. This budget supports the creation of simulated financial records, tax strategy scenarios, and behavioral datasets used for training and evaluating the recommendation models.

Testing and Prototype Demonstration Expenses

- This cost supports the development of demonstration dashboards, visualization tools, and evaluation reports used to showcase the functionality and performance of the proposed system. These resources are necessary to present the research outcomes during proposal evaluations, presentations, and final demonstrations.

Contingency Allocation

- A contingency allocation is included to address unexpected expenses that may arise during the research process, such as additional cloud compute usage, extended data storage requirements, or additional testing resources.

7.3 Budget Feasibility

The proposed budget is moderate and feasible for a machine learning-based research project. Unlike projects that require specialized hardware equipment, this research mainly depends on software development and computing resources.
The allocated budget is sufficient to support model development, experimentation, dataset preparation, and prototype demonstration, while maintaining cost efficiency.

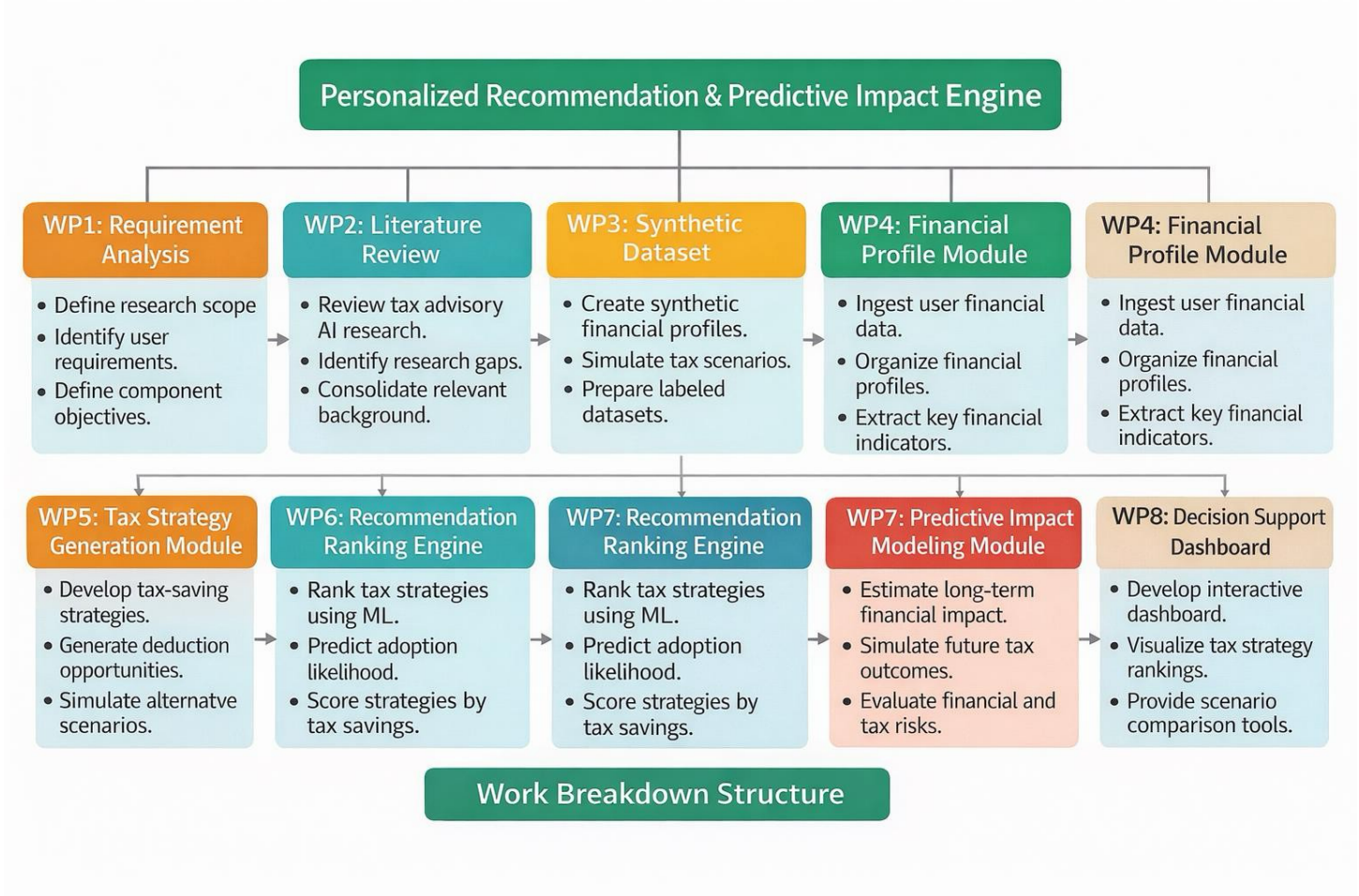
7.4 Cost Optimization Considerations

To ensure efficient use of financial resources, the project will prioritize the following strategies:

• Using open-source machine learning libraries and development frameworks
• Utilizing free academic tools and research resources where available
• Generating synthetic datasets instead of collecting expensive real-world financial data
• Using cloud computing only when necessary for model training and simulations

These strategies help maintain a realistic and cost-effective budget while ensuring that the research achieves its technical objectives.

8. WORK BREAKDOWN STRUCTURE (WBS)



WP1: Requirement Analysis and Research Planning

This work package is concerned with establishing the research scope, research problem, and research objectives and requirements for the Personalized Recommendation and Predictive Impact Modeling Engine. This includes the analysis of the needs of taxpayers and financial advisory systems in order to ascertain how a personalized recommendation for taxes can be achieved. This stage is essential in ensuring that the research component is in line with the overall project goals and that the architecture and plan for the system are clear before the development of the system begins.

WP2: Literature Review and Research Gap Analysis

This work package includes the review of existing research literature, tax advisory systems, and applications of machine learning technology in the field of financial decision-support systems. The review is mainly focused on tax recommendation systems, learning to rank algorithms, predictive financial models, behavioral adoption predictions, and financial forecasting techniques.

The main objective of this phase is to identify the limitations of existing systems and the research gap filled by the proposed system. This work package lays the theoretical basis for the research.

WP3: Synthetic Financial Dataset Construction

This work package focuses on creating and preparing datasets required for training and evaluating the machine learning models used in the system. Since real taxpayer data is confidential, synthetic datasets will be generated to simulate realistic financial profiles and tax planning scenarios. This includes generating simulated user income records, tax deduction scenarios, financial behavior indicators, and strategy adoption patterns. The dataset will then be structured and divided into training, validation, and testing datasets to support model development and experimental evaluation.

WP4: User Financial Profile Module Development

This work package focuses on developing the module responsible for collecting, organizing, and processing user financial information. The module will ingest financial data such as income level, salary history, deductions, and financial preferences. It will also extract important financial indicators such as disposable income, tax eligibility factors, and risk tolerance indicators. This structured financial profile will serve as the primary input for generating tax strategies and recommendation models.

WP5: Tax Strategy Generation Module Development

This work package focuses on developing the module that generates potential tax-saving strategies based on Sri Lankan tax regulations and deduction rules. The system will create multiple candidate strategies by analyzing available tax benefits, allowable deductions, and financial eligibility conditions. These strategies may include adjustments to deductions, optimized financial allocations, or alternative tax planning approaches. The generated strategies will serve as input candidates for the recommendation ranking system.

WP6: Recommendation Ranking Engine Development

This work package involves implementing the machine learning-based recommendation ranking model that prioritizes tax strategies according to their suitability for the user. The system will use learning-to-rank algorithms to evaluate each strategy based on factors such as expected tax savings, feasibility, financial risk, and predicted adoption likelihood. The goal of this module is to ensure that the most relevant and practical strategies appear at the top of the recommendation list.

WP7: Predictive Impact Modeling Module Development

This work package focuses on developing the predictive modeling engine that estimates the long-term financial consequences of adopting different tax strategies. The system will implement income projection models, tax liability forecasting, and simulation techniques such as Monte Carlo simulation to analyze possible future financial outcomes. This module allows users to understand how their current tax decisions may influence future income, tax liabilities, and financial risks.

WP8: Evaluation and Experimental Validation

This work package ensures the scientific evaluation and validation of the proposed system. The developed models will be tested using simulated financial datasets and predefined evaluation metrics. The evaluation process will analyze recommendation accuracy, prediction accuracy, ranking effectiveness, and financial outcome estimation reliability. Experimental validation will also identify potential limitations and opportunities for improvement in the recommendation and prediction models.

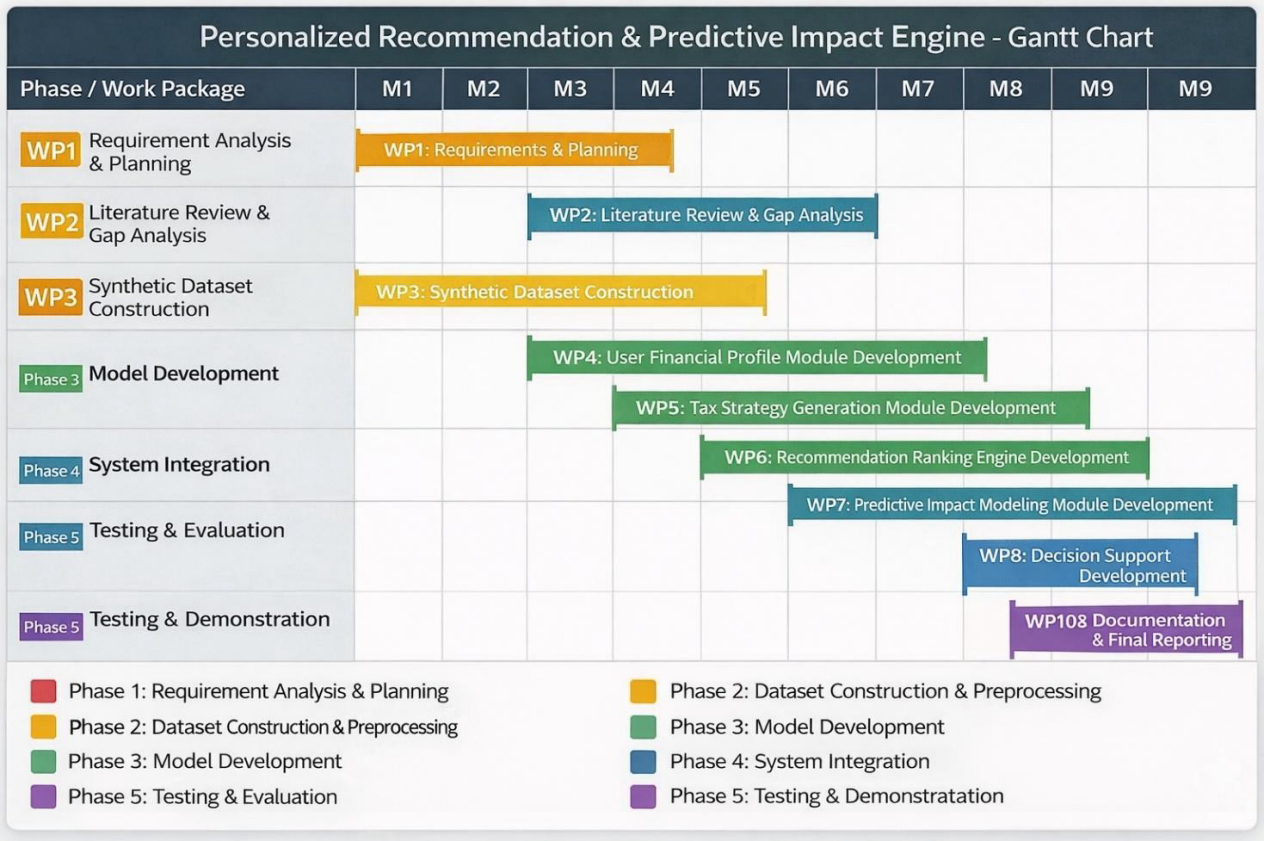
WP9: Prototype Integration and Demonstration

This work package focuses on integrating the developed modules into a working prototype system. The modules including the financial profile module, tax strategy generator, recommendation ranking engine, predictive impact model, and visualization dashboard will be connected into a single workflow. Demonstration scenarios will be prepared to illustrate how the system generates personalized tax recommendations and predicts future financial outcomes for users.

WP10: Documentation and Final Reporting

This final work package covers the preparation of the research documentation and final deliverables. It includes writing the final research report, documenting system architecture and methodology, summarizing experimental results, and preparing presentation materials. This stage ensures that all research findings, technical contributions, and implementation details are clearly documented and communicated for evaluation and future reference.

9. GANTT CHART



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